Freight Rate Prediction Assessment

Generated with seed 42 on 2026-08-01T00:10:19.912483+00:00 using Python 3.12.10. Git commit: not a git repository.

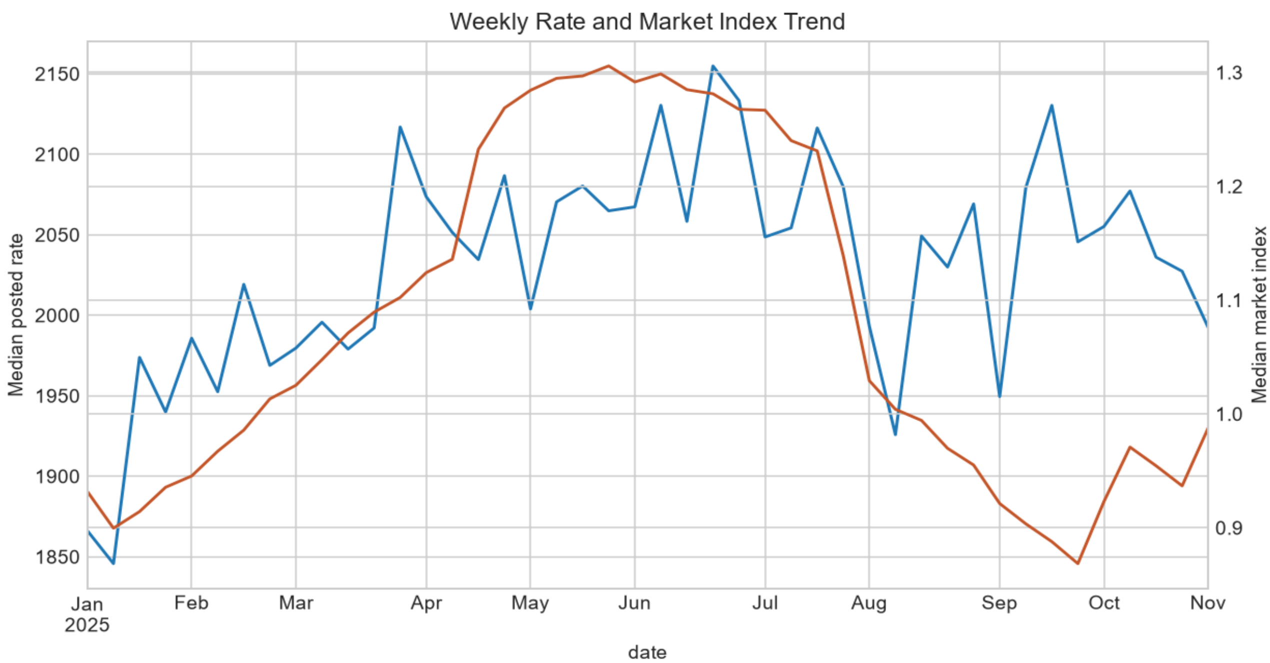
Executive Summary

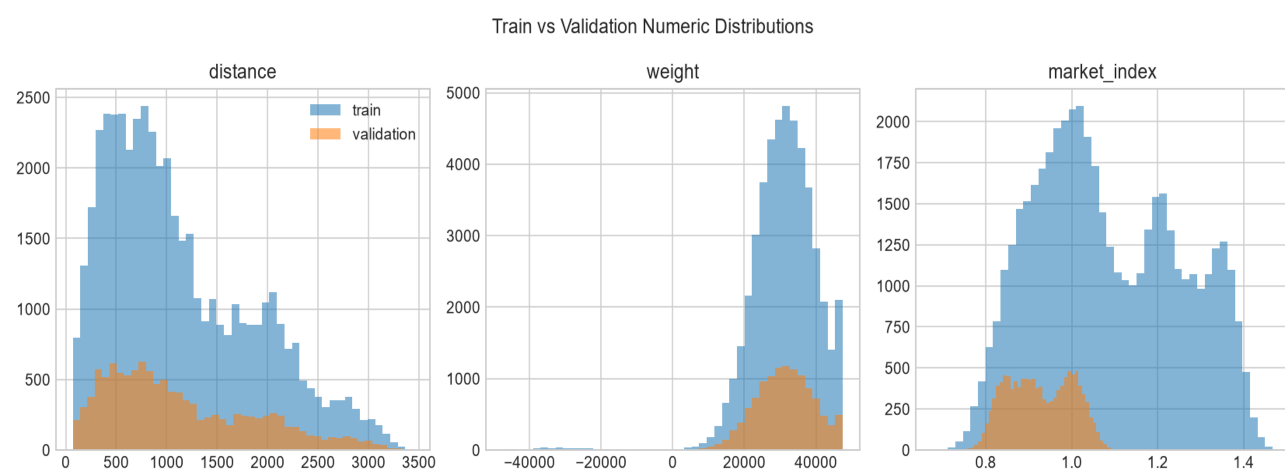
The final full-feature model is hgb_mae_full_abs_weight selected by mean out-of-time MAE. The fixed December chart uses december_hgb_mae_abs_weight, a separate model trained without market_index or quote_signal because those inputs are absent from the chart file. No hidden validation target values were used or inferred.

Data Overview and Quality

Training data contains 48,000 rows from 2025-01-01 through 2025-10-31; validation contains 12,000 rows from 2025-11-01 through 2025-12-31. Training has 300 missing weights, 374 missing market_index values, and 292 negative weights.

Validation includes 8 cities not observed in labeled training and 736 unseen exact pickup-delivery routes. Market index mean shifts from 1.083 to 0.927.





Validation Design

Model selection uses chronological expanding-window validation. Each fold trains only on earlier dates than its validation period, mirroring the final future prediction task.

fold	train_start	train_end	valid_start	valid_end	train_rows	valid_rows
validate_july	2025-01-01	2025-06-30	2025-07-01	2025-07-31	28,806.000	4,912.000
validate_august	2025-01-01	2025-07-31	2025-08-01	2025-08-31	33,718.000	4,759.000
validate_september	2025-01-01	2025-08-31	2025-09-01	2025-09-30	38,477.000	4,670.000
validate_october	2025-01-01	2025-09-30	2025-10-01	2025-10-31	43,147.000	4,853.000
validate_sep_oct	2025-01-01	2025-08-31	2025-09-01	2025-10-31	38,477.000	9,523.000

Feature Engineering and Treatments

Features include date seasonality, geography from coordinates, haversine distance, route and city-pair categories, frequency features fitted inside each training fold, equipment, distance, weight flags, and market/quote interactions for the full model. Negative weights were benchmarked as absolute-value repair and as missing-with-flag; missing market_index values are imputed from fold training medians with a missingness flag.

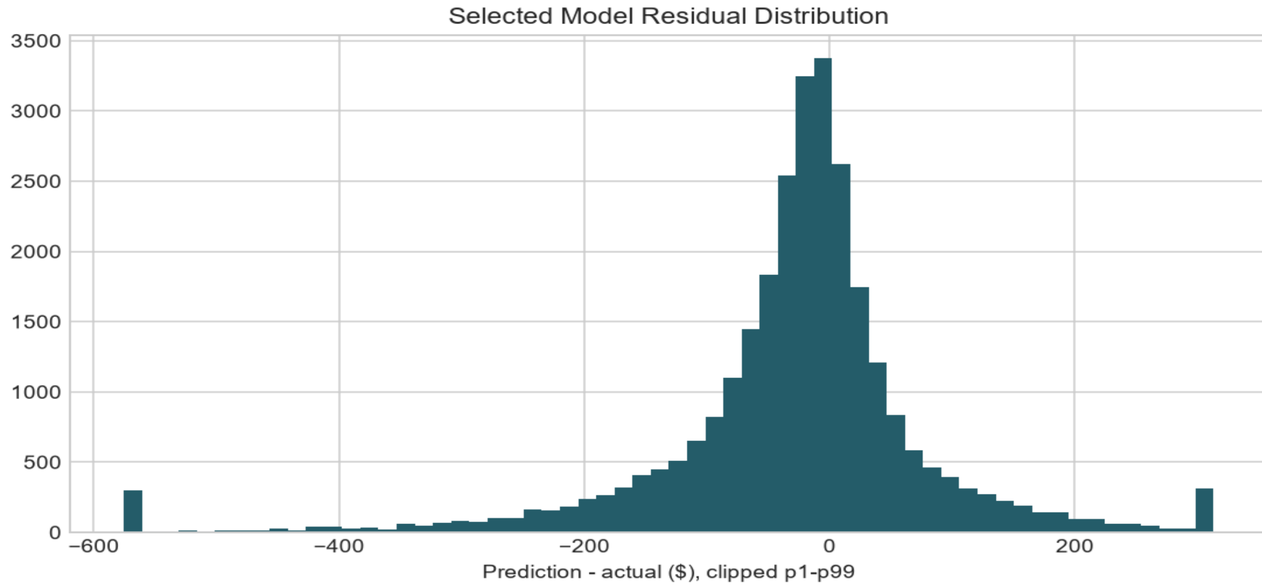
Model Comparison

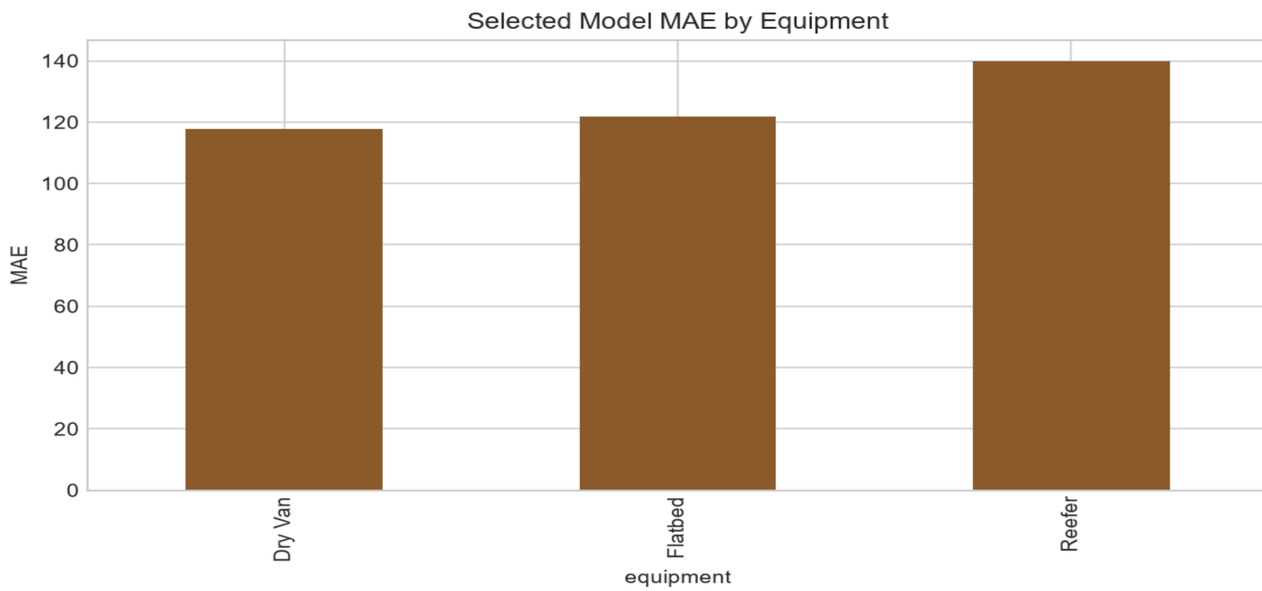
Baselines include global median, distance-only linear regression, equipment median rate-per-mile, and ridge regression. Nonlinear candidates include histogram gradient boosting and extra trees with controlled deterministic settings.

model	mean_mae	recent_mae	mean_rmse	mean_r2	mean_wape
hgb_mae_full_abs_weight	123.973	121.005	633.276	0.824	0.052
hgb_mae_full_missing_bad_weight	124.281	122.155	634.024	0.824	0.052
extra_trees_full_abs_weight	164.837	161.999	653.291	0.813	0.069
ridge_log_full_abs_weight	185.254	169.559	659.887	0.809	0.078
distance_linear	194.338	192.774	650.237	0.815	0.081
equipment_median_rpm	227.685	231.305	668.086	0.805	0.095
global_median	1,140.511	1,146.794	1,553.975	-0.057	0.478

Error Analysis and Limitations

The primary metric is MAE because posted rates are strongly right-skewed with rare high-price tails that can dominate RMSE. RMSE, WAPE, seen/unseen-route MAE, and normal/extreme target subsets were retained as diagnostics. The main limitation is that final validation labels are hidden, so all model decisions rely on labeled temporal backtests only.



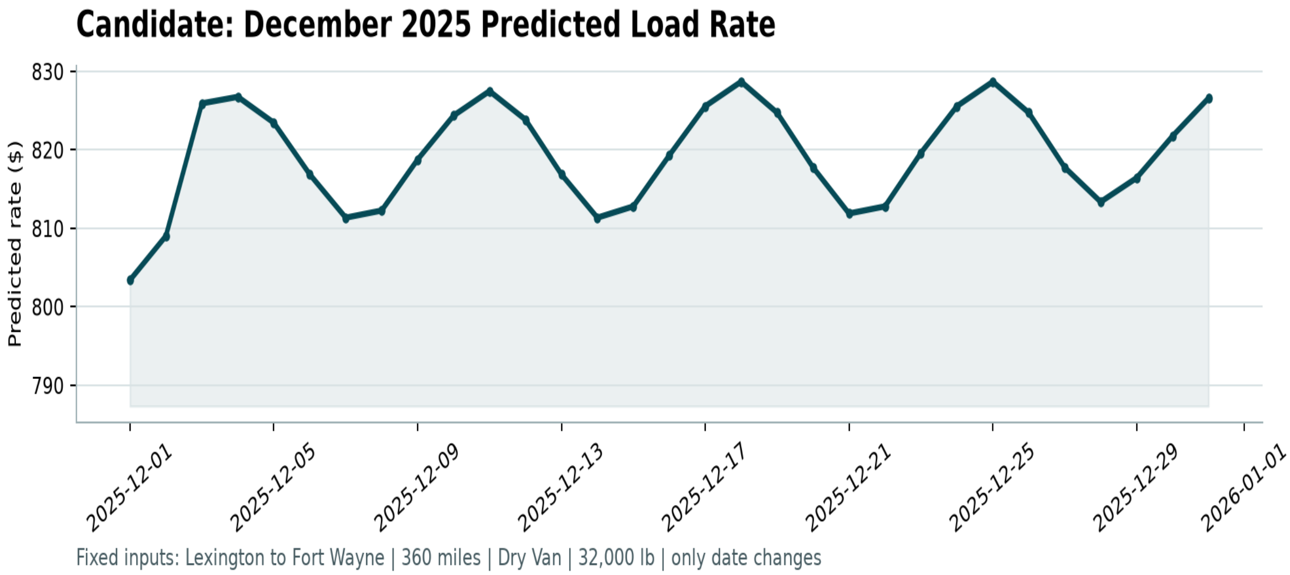


Fixed December Prediction

The December chart input lacks exogenous market fields, so model choice for this file used reduced-feature temporal backtesting under the same missing-feature contract.

model	mean_temporal_mae
december_hgb_mae_abs_weight	115.887
december_extra_trees_abs_weight	175.179
december_ridge_log_abs_weight	212.892
december_equipment_median_rpm	227.685

The following image is the exact chart generated by the supplied score.py from the completed December prediction CSV.



Reproducibility

From a clean checkout, install requirements and run: python scripts/run_pipeline.py --project-root . Then validate with: python scripts/validate_outputs.py --project-root . and python score.py --predictions validation_predictions.csv --december-predictions december_chart_inputs.csv.

Human-only remaining steps are publishing the GitHub repository, recording the Loom walkthrough, pasting the Loom URL, and submitting the assessment.